Simpson's Paradox Shuffle Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	**Time**	35-45 min
Difficulty	Medium	**ID**	simpsons-paradox-shuffle

Big idea: Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

Overview

Students sort subgroup data cards and discover that an overall trend can reverse when data is grouped differently.

Student Prompt

Inspect Simpson's Paradox Shuffle Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Simpson's Paradox Shuffle Stimulus A	Student-facing stimulus 1: Method A beats Method B overall, but Method B performs better inside each difficulty subgroup. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Simpson's Paradox Shuffle Stimulus B	Student-facing stimulus 2: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Simpson's Paradox Shuffle Reveal / Twist	Student-facing stimulus 3: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Simpson's Paradox Shuffle Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Simpson's Paradox Shuffle Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.

#	Question for students	Teacher key / cue
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	How do grouping choices shape statistical knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

• Use small whole-number subgroups so the reversal is visible without calculators.

• Ask students for the headline claim before revealing subgroup structure.

Ready-to-Use Examples

• Method A beats Method B overall, but Method B performs better inside each difficulty subgroup.

• Headline card: 'Method A has the higher success rate' beside subgroup table cards.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups cards for two methods split across two subgroups.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students calculate or compare success rates inside each subgroup.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups combine the cards and calculate the overall rate.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Reveal the reversal and ask which claim is more responsible.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why a mathematically accurate headline can still mislead.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

• The calculation is correct; now ask whether the comparison is fair.

• What vanished when the data was combined?

Common Misconceptions

• Students may treat a persuasive pattern as equivalent to proof.

• Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

• Distinguishes intuition, empirical pattern, model, and proof.

• Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

• Support: provide completed percentages and focus on interpretation.

• Challenge: ask students to design the smallest dataset that creates a reversal.

Extension Tasks

• Students create a simple fictional dataset where aggregation changes the apparent story.

• Connect the paradox to admissions, medicine, sport, or school data.

Related Activities

• Correlation Illusion Cards

• Average Trap

• P-hacking with Random Data